



SYMBIOSIS INSTITUTE OF TECHNOLOGY (SIT)

Constituent of Symbiosis International (Deemed University), Pune

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Re-Accredited by NAAC with 'A++' Grade

ASSIGNMENT No: 11

TITLE: Write a Java program to demonstrate the implementation of interfaces.

Problem Statement

Write a Java program to demonstrate the implementation of interfaces.

Learning Objectives

To implement interfaces for achieving abstraction and multiple inheritance of type.

Theory

An interface specifies a set of abstract methods that implementing classes must define. It provides complete abstraction and supports multiple inheritance of type. Interfaces improve modularity, loose coupling, and software flexibility. They allow different classes to implement common behavior while maintaining independent implementations. Interfaces are widely used in enterprise application development.

Conclusion:

The Java program was successfully implemented to demonstrate the use of interfaces. Interfaces were used to achieve abstraction by defining common methods that were implemented differently by multiple classes. The Printable and Switchable examples showed how interfaces support code reusability, loose coupling, and multiple inheritance of type. This experiment enhanced the understanding of object-oriented programming concepts and the practical use of interfaces in Java.

Exercise

1. Create an interface Printable and implement it in Student and Employee classes.

Code:

```
1 interface Printable {
2     void print();
3 }
4
5 class Student implements Printable {
6     String name;
7     int rollNo;
8
9     Student(String name, int rollNo) {
10         this.name = name;
11         this.rollNo = rollNo;
12     }
13
14     public void print() {
15         System.out.println("Student Name: " + name);
16         System.out.println("Roll No: " + rollNo);
17     }
18 }
19
20 class Employee implements Printable {
21     String name;
22     int id;
23
24     Employee(String name, int id) {
25         this.name = name;
26         this.id = id;
27     }
28
29     public void print() {
30         System.out.println("Employee Name: " + name);
31         System.out.println("Employee ID: " + id);
32     }
33 }
34
35 class Main {
36     public static void main(String[] args) {
37         Printable p1 = new Student("Anurag Thakur", 22198);
38         Printable p2 = new Employee("Rahul", 101);
39
40         p1.print();
41         p2.print();
42     }
43 }
```

Output:

```
Student Name: Anurag Thakur
Roll No: 22198
Employee Name: Rahul
Employee ID: 101

=== Code Execution Successful ===
```

2. Create an interface Switchable with a method turnOn(). Implement the interface in Light and Fan classes to display the device status.

```
interface Switchable {  
    void turnOn();  
}
```

```
class Light implements Switchable {  
    public void turnOn() {  
        System.out.println("Light is ON");  
    }  
}
```

```
class Fan implements Switchable {  
    public void turnOn() {  
        System.out.println("Fan is ON");  
    }  
}
```

```
class Main {  
    public static void main(String[] args) {  
        Switchable s1 = new Light();  
        Switchable s2 = new Fan();  
  
        s1.turnOn();  
        s2.turnOn();  
    }  
}
```

```
}  
}
```

OUTPUT

```
Output  
Light is ON  
Fan is ON
```